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Gated Cross-Modal Fusion and Feature Reconstruction for Visible-Infrared Remote Sensing Object Detection

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***Abstract*—The visible–infrared object detection in UAV remote sensing remains challenging due to small object feature degradation, complex background interference, modality heterogeneity, and the high computational cost of existing fusion methods. To address these issues, this study proposes a multimodal gated interaction and reconstruction network (MGIR-Net). First, a feature reconstruction module (FRM) recalibrates intra-modal representations by enhancing local saliency, suppressing directional background interference, and compensating global response bias. Second, a cross-modal gated interaction module (CGIM) performs GELU-based gated modulation on fused visible-infrared features to enhance reliable complementary cues while suppressing unreliable responses. Third, a hierarchical contextual guidance module (HCGM) enlarges the effective receptive field through decoupled spatial sampling, providing spatial priors for small object localization with limited overhead. The effectiveness and generalization ability of MGIR-Net are demonstrated by extensive experiments on two aerial remote sensing datasets, DroneVehicle and VEDAI, and four complementary visible–infrared benchmarks, LLVIP, M^{3}FD, MFAD, and KAIST. The proposed method achieves mAP50 scores of 85.5%, 78.6%, 96.5%, 90.5%, 79.0%, and 75.2% on the six datasets, respectively. With 6.9M parameters and 19.3 GFLOPs, MGIR-Net provides a favorable balance between detection accuracy and computational efficiency. The source code for this work is publicly available at https://github.com/Leon1998314/MGIR-Net.**

***Index Terms*—UAV aerial images, small object detection, visible-infrared, cross-modal gated interaction module (CGIM), feature reconstruction module (FRM)**

I. INTRODUCTION

THE rapid advancement of unmanned aerial vehicle (UAV) technology has ushered in a transformative era in remote sensing. Leveraging high-mobility, on-demand deployment, and the capacity for ultra-high spatial resolution data acquisition, UAVs offer unique operational advantages [1], [2]. By integrating various sophisticated airborne sensors, these platforms have become an essential component of contemporary multidimensional sensing architectures [3]. The utility of UAV applications is evident in time-critical scenarios, such as intelligent transportation systems and real-time disaster response [4, 5]. Within these demanding contexts, UAVs have evolved from basic data collection tools into intelligent distributed edge-computing nodes, enabling real-time information aggregation and autonomous mission execution.

In recent years, object detection algorithms have transitioned from classical methodologies to deep learning paradigms, with convolutional neural networks (CNNs) emerging as the dominant architecture [6]. While the pioneering R-CNN series [7], [8] established high-precision benchmarks, subsequent frameworks such as the YOLO series [9], [10], [11] and SSD [12] prioritized inference speed to satisfy real-time requirements. Despite their proficiency in capturing local features, CNN-based approaches exhibit inherent limitations: their constrained receptive fields impede the comprehension of global contextual information, and they typically rely on manual post-processing heuristics, such as non-maximum suppression (NMS), which involve complex parameter tuning. To circumvent these challenges, Transformer-based models, notably the DETR series [13], [14], [15] have emerged as a robust alternative. By leveraging self-attention mechanisms, Transformers model long-range global dependencies across the entire image, demonstrating promising performance in dense environments and overlapping target scenarios where traditional CNNs frequently falter. Nevertheless, applying these general frameworks to UAV-based remote sensing imagery remains arduous due to high-density small objects that are susceptible to background interference and information loss during successive downsampling operations [16], [17].

To overcome unimodal limitations, integrating visible and infrared modalities has become a prominent research direction, as illustrated in Fig. 1. However, this fusion paradigm faces three core challenges. First, simple concatenation, addition, or fixed attention-based fusion often provides insufficient cross-modal interaction, fails to fully exploit complementary information, and may introduce redundant features. Second, substantial differences in imaging mechanisms, texture characteristics, and noise distributions between visible and

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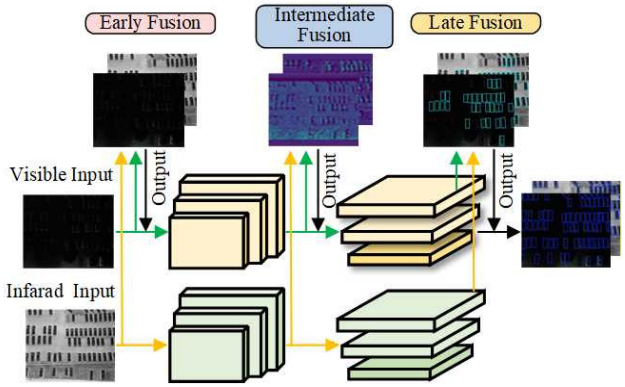


Fig. 1. Diagram of different fusion strategies.

infrared modalities make it difficult to dynamically suppress unreliable modal responses and balance their contributions. Third, after multimodal fusion, small object responses can be overwhelmed by enhanced background features, leading to weakened target details and boundaries, while existing methods provide limited reconstruction mechanisms to recover these degraded representations. In addition, increasingly complex attention architectures often impose considerable computational costs, hindering real-time deployment on embedded UAV platforms. Therefore, a robust and efficient framework is highly desirable to enable adaptive cross-modal interaction, reduce background interference [18], and achieve a better trade-off between detection accuracy and computational efficiency in complex environments [19].

Existing methods typically combine visible and infrared features through fixed operations or a single attention mechanism, which may limit their ability to adaptively model modality-specific reliability and cross-modal complementarity. Moreover, post-fusion processing often relies on generic convolutional or attention-based refinement, making it difficult to preserve weak small object cues and compensate for information attenuation caused by multimodal fusion. To address these limitations, this study proposes a lightweight multimodal gated interaction and reconstruction network (MGIR-Net). Specifically, FRM adopts three parallel branches: local saliency reconstruction (LSR), directional semantic calibration (DSC), and global bias compensation (GBC), followed by residual feature aggregation. CGIM consists of point-wise projection, depth-wise convolution, dual-branch GELU-based modulation, residual connection, and MLP refinement. HCGM combines local and dilated depth-wise sampling branches with hierarchical kernel configurations, followed by feature concatenation and point-wise fusion. The primary contributions are:

1) We propose MGIR-Net, a lightweight visible-infrared framework for small object detection in remote sensing images. The network integrates FRM, CGIM, and HCGM into a unified architecture.
2) FRM combines LSR, DSC, and GBC to reconstruct modality-specific and fused features, helping preserve small object details and reduce background interference.
3) CGIM employs dual-branch gated modulation guided by joint visible-infrared features and a visible-feature residual path to preserve structural details, thereby reducing background interference and feature attenuation.
4) HCGM uses hierarchical decoupled spatial sampling to capture multi-scale contextual information and provide spatial priors for small object detection.
5) Experiments on DroneVehicle, VEDAI, LLVIP, M³FD, MFAD, and KAIST demonstrate the competitive accuracy, robust applicability across different scenarios, and computational efficiency of MGIR-Net.

The remainder of this paper is organized as follows: Section II reviews related work on deep-learning-based small object detection and visible-infrared multimodal detection. Section III elaborates on the proposed MGIR-Net architecture. Section IV presents experimental results and a comparative analysis with state-of-the-art methods. Section V concludes the paper.

II. Related Work

A. Small Object Detection

Despite recent advances in deep learning, small object detection remains challenging because spatial details and semantic information gradually diminish during feature extraction. To mitigate this loss of features, contemporary research has primarily focused on two directions: first, constructing robust multi-scale representations to preserve spatial fidelity [20]; and second, exploiting contextual information to derive relational cues from the surrounding environment.

The core of multi-scale representation lies in integrating low-level structural cues with high-level semantics to maintain the feature integrity of small objects. Recent studies on hyperspectral imaging further emphasize the importance of rich spectral information for preserving subtle target characteristics when spatial details are limited [21]. Lin et al. [22] pioneered the FPN, which establishes a feature pyramid via top-down paths and lateral connections. RetinaNet [23] combined this with Focal Loss to address the extreme foreground-background imbalance in multi-scale fusion. RoI Transformer [24] introduced adaptive rotated proposals, while ReDet [25] employed rotation-equivariant convolutions to fundamentally enhance feature extraction for oriented objects: Duan et al. [26] designed Focal Spotter to amplify small object features via a focusing module. Regarding efficiency and robustness, Li et al. [27] developed lightweight pyramids to maintain semantic consistency on resource-constrained platforms. For scene-specific optimizations, Yuan et al. [28]implemented denoising using combined column and spatial attention. More recently, SpectralGPT [29] explored large-scale pretraining for transferable spectral representations, providing a new perspective for capturing discriminative cues in complex remote sensing scenes.

Beyond fine-grained feature extraction, leveraging environmental context has become a vital strategy to compensate for single-representation limitations. Zhang et al. [30] introduced dynamic local context tables to adapt to target

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locality and scale variations, while Yang et al. [31] proposed ClusDet to explicitly model spatial distributions by unifying clustering and detection.

Contemporary research increasingly emphasizes the integration of global associations. For instance, Zhang et al. [32] proposed FFCA-YOLO, which utilizes spatial context awareness to suppress background noise; Du et al. [33] introduced CEASC to balance global enhancement and efficiency via adaptive sparse convolutions; and Chen et al. [34] achieved texture alignment through global-local fusion and semantic guidance. Furthermore, Hu et al. [35] captured global features in the frequency domain, while Huang et al. [36] proposed BRSTD, introducing biomimetic antagonistic receptive fields to enhance the contrast between small objects and their environment. LRR-Net [37] incorporates low-rank representation priors into a deep unfolding network for hyperspectral anomaly detection, demonstrating the value of combining structural priors with deep feature learning. Existing studies attempt to enhance small object features through global context but often overlook the semantic interference and feature degradation caused by salient backgrounds. Due to insufficient consideration for feature distribution patterns, these models struggle to execute directional semantic isolation to extract distinct small object representations. Additionally, frequent downsampling within deep multi-scale feature pyramids leads to the gradual dilution of small object features by background pixels. Meanwhile, conventional large-kernel convolution schemes remain computationally expensive for real-time embedded deployment.

B. Visible-Infrared Fusion Object Detection

The early research on visible-infrared fusion object detection typically formulated image fusion and object detection as two decoupled stages [38], [39]. This isolation often leads to optimization conflicts, where texture preservation in the fused image might not align with the semantic requirements for precise object localization [40], [41]. To address this discrepancy, recent methods have focused on joint frameworks that tightly couple fusion and detection, ensuring that the extracted cross-modal features are inherently optimized for the detection task [42], [43] .

Under this collaborative framework, Xu et al. [44] proposed U2Fusion, which achieves adaptive fusion for multimodal and multi-exposure images by preserving salient source information. Zeng et al. [45] introduced MMI-Det, which integrates complementary infrared and visible information via contour enhancement, fusion focusing, and contrast bridging. Chen et al. [46] proposed GLFDet, which uses global and local optimization modules to assess input quality in the frequency domain and select informative channel features. Transformers are widely used for their strong contextual modeling capabilities. For instance, Yuan et al. [47] proposed C²Former, leveraging cross-attention for feature fusion and alignment. CFT [48] employs a Transformer structure to model global interactions, while Shen et al. [49] introduced ICAFusion, utilizing a dual cross-attention structure to simultaneously model inter-modal complementary information. From a frequency perspective, Chen et al. [50] proposed MGFF, which employs wavelet transforms to enhance modal-specific features. More recently, Mamba-based architectures have provided a new paradigm for cross-modal fusion. Zhang et al. [51] introduced MSTD, using mask-enhanced pixel-level fusion to learn semantically aligned representations. Zhou et al. [52] further developed the DMM framework with a disparity-guided cross-modal fusion Mamba module for adaptive feature integration. Wang et al. [53] proposed MGMF, combining mask regularization with Mamba-based state-space fusion to reduce modal discrepancies and improve shared feature representations across modalities. Recent research has advanced cross-modal interaction in multimodal remote sensing detection. CMFADet [54] introduces adaptive cross-modal interaction for dynamic feature fusion and deep complementarity. LCAFNet [55] employs visible and infrared modalities as mutual guidance to iteratively extract complementary representations through cross-attention. MCISFNet [56] combines saliency-guided soft fusion with multi-scale interactive gating to compensate for missing modality information and enhance fine-grained features. CCSFuse [57] adopts collaborative feature compensation and selective fusion to strengthen inter-modal complementarity while reducing modal conflicts. PAO-YOLO [58] addresses parallax interference and small object detection through modality-invariant feature disentanglement and deformation-adaptive fusion.

In summary, traditional direct fusion methods, such as concatenation or simple weighting, are vulnerable to environmental factors like shadow obscuration and local occlusion in infrared and visible modalities. When one modality provides spurious or blurred information, it can lead to semantic contamination of the fused features. In the absence of effective target response modulation, cross-modal feature mismatch further limits the discriminability of small objects within deep networks. Although mainstream methods, such as Transformer or Mamba, attempt to introduce complex long-range modeling mechanisms to enhance performance [59] , [60], they incur substantial computational overhead and inference latency. Furthermore, global attention mechanisms tend to amplify both target and background features simultaneously when processing extremely small pixels, making it challenging for models to simultaneously maintain computational efficiency and detection precision.

III. PROPOSED METHOD

To address the challenges in multimodal detection for aerial remote sensing, specifically feature degradation from redundant background features, semantic confusion in cross-modal fusion, and spatial detail dissipation in deep propagation, this paper proposes MGIR-Net, a network for multimodal perception and fusion as illustrated in Fig. 2. Building upon the foundational C3k2, SPPF, and C2PSA from YOLO11, MGIR-Net incorporates a dual-stream backbone for RGB and IR modalities. During dual-modal feature extraction, the FRM is embedded within C3k2 structures and is also applied after cross-modal

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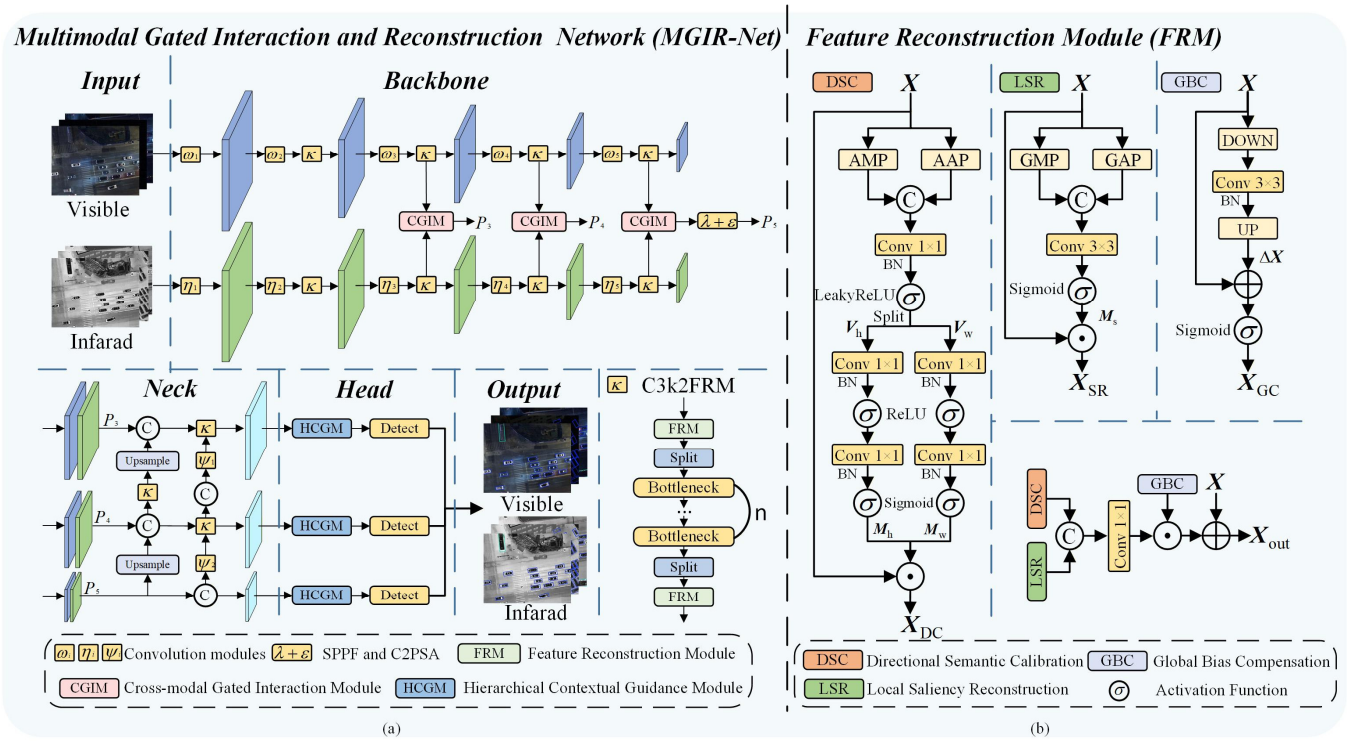


Fig. 2. Overview of the proposed model. (a) Overall architecture of the proposed method MGIR-Net. (b) Architecture of the FRM.

fusion. In the dual-branch feature fusion stage, CGIM is inserted at the corresponding fusion nodes of the visible and infrared branches. Finally, HCGM is placed before the detection head to perform decoupled spatial sampling on hierarchical feature maps.

A. Feature Reconstruction Module

In the domain of visible-infrared cross-modal object detection, the detection accuracy of small objects is persistently hindered by salient background features and irrelevant semantic interference. Such disturbances frequently result in the feature representation being dominated by redundant background responses, leading to the feature degradation of small objects within complex environments. To mitigate these adverse effects, this study introduces the feature reconstruction module (FRM), as illustrated in Fig. 2 (b). By enhancing the saliency representation of small objects, directionally stripping irrelevant semantic interference, and implementing global bias compensation, the FRM performs feature-distribution calibration. This mechanism enhances detection accuracy for small objects in complex backgrounds.

Prior to entering the FRM, input features undergo preliminary preprocessing, utilizing 3×3 convolutions and LeakyReLU activation functions to facilitate initial spatial feature aggregation, thereby yielding more discriminative input representations for subsequent stages. The FRM comprises three interconnected components: LSR, DSC, and GBC. By employing a parallel processing framework, the module supports improved detection of small objects.

Although FRM incorporates several operations commonly used in attention mechanisms, such as global pooling, directional pooling, and sigmoid-based feature reweighting, its overall structure differs from those of CBAM, Coordinate Attention (CA) [61], and GLU [62]. CBAM [63] sequentially applies channel attention and spatial attention, whereas FRM adopts three complementary branches, namely LSR, DSC, and GBC. CA mainly encodes positional information through separate one-dimensional pooling along the horizontal and vertical directions. In contrast, directional modeling in DSC is only one component of FRM and is further combined with local saliency weighting and global bias correction. GLU generally splits the input features into a value branch and a gate branch, followed by element-wise multiplication. Unlike GLU, FRM does not rely on a single value-gate splitting operation; instead, it reconstructs features through the coordinated use of saliency weighting, directional calibration, global bias correction, and residual feature fusion. Therefore, FRM should be regarded as a multi-branch feature reconstruction block rather than a direct variant of a single attention or gating module.

In the LSR module addressing the characteristic that small objects occupy an extremely small proportion of pixels in the spatial dimension, the module concurrently employs global average pooling (GAP) and global max pooling (GMP) to extract global contextual features, thereby generating the feature saliency weight $\boldsymbol{M}_{\mathrm{s}}$, enabling small object responses to achieve high representational saliency within complex environmental backgrounds:

$$\boldsymbol{M}_{\mathrm{s}} = \sigma_{\mathrm{sig}}\left(\mathcal{F}_{3\times 3}[\mathrm{GAP}(\boldsymbol{X}), \mathrm{GMP}(\boldsymbol{X})]\right) \tag{1}$$

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$$\boldsymbol{X}_{\mathrm{SR}} = \boldsymbol{X} \odot \boldsymbol{M}_{\mathrm{s}} \tag{2}$$

where σ_{sig} denotes the Sigmoid activation function, $\mathcal{F}(\cdot)$ represents a convolutional operation with a specific kernel size, $\odot$ denotes the Hadamard product, and $[\cdot,\cdot]$ denotes the concatenation operation.

In the DSC module, addressing the issue of irrelevant semantic interference caused by complex background interference, the feature map is projected into a decoupled vector space via one-dimensional adaptive maximum pooling (AMP) and adaptive average pooling (AAP), facilitating efficient modeling of feature distributions in both horizontal and vertical directions. This operation obtains the response distribution differences between the horizontal and vertical dimensions, constructing a directional descriptor $\boldsymbol{V}_{\mathrm{d}}$ that can characterize background distribution patterns through a concatenation operation. Subsequently, a split operation is utilized to decouple the fused features along the channel dimension, yielding the vertical descriptor $\boldsymbol{V}_{\mathrm{h}}$ and the horizontal descriptor $\boldsymbol{V}_{\mathrm{w}}$, respectively. Through successive linear projections and nonlinear mappings, a deep reconstruction of the feature space is achieved, ultimately generating adaptive weights $\boldsymbol{M}_{\mathrm{h}}$ and $\boldsymbol{M}_{\mathrm{w}}$. These weights enable the precise suppression of dominant interference responses in specific orientations. Finally, directional dimensional calibration is performed on the original features using the generated adaptive weights, concurrently filtering redundant background information and enhancing target feature intensity to obtain the interference-free semantic feature map $\boldsymbol{X}_{\mathrm{DC}}$.

$$\boldsymbol{V}_{\mathrm{d}} = \sigma_{\mathrm{L}}(\mathrm{BN}(\mathcal{F}_{1\times1}[\mathrm{AAP}(\boldsymbol{X}), \mathrm{AMP}(\boldsymbol{X})])) \tag{3}$$

$$\boldsymbol{V}_{\mathrm{h}}, \boldsymbol{V}_{\mathrm{w}} = \mathrm{Split}(\boldsymbol{V}_{\mathrm{d}}) \tag{4}$$

$$\boldsymbol{M}_{\mathrm{h}} = \sigma_{\mathrm{sig}}(\mathcal{F}_{1\times1}(\delta(\mathcal{F}_{1\times1}(\boldsymbol{V}_{\mathrm{h}})))) \tag{5}$$

$$\boldsymbol{M}_{\mathrm{w}} = \sigma_{\mathrm{sig}}(\mathcal{F}_{1\times1}(\delta(\mathcal{F}_{1\times1}(\boldsymbol{V}_{\mathrm{w}})))) \tag{6}$$

$$\boldsymbol{X}_{\mathrm{DC}} = \boldsymbol{X} \odot \boldsymbol{M}_{\mathrm{h}} \odot \boldsymbol{M}_{\mathrm{w}} \tag{7}$$

where σ_{L} denotes the LeakyReLU activation function, δ represents the ReLU activation function, and BN stands for the batch normalization layer. The Split denotes the partitioning of the feature map along the channel dimension.

By obtaining the distribution patterns along the axial dimensions, this mechanism achieves the directional stripping of incoherent semantic interference. While reducing the wrong detection rate, it helps preserve the local saliency of small objects within complex backgrounds.

The GBC is designed to perform global compensation on features through an asymmetric sampling mechanism. By utilizing downsampling operators to significantly expand the receptive field, it captures the global distribution prior of the background, effectively establishing a robust baseline for background characterization. By re-projecting this prior information back to the original spatial scale, the module generates a spatially continuous correction amount $\Delta\boldsymbol{X}$. By superimposing $\Delta\boldsymbol{X}$ onto the original feature map $\boldsymbol{X}$, the module achieves dynamic bias correction of the feature responses. This process accurately offsets spatial distribution artifacts caused by hardware response non-uniformity, thereby promoting more stable spatial responses at the global scale:

$$\Delta\boldsymbol{X} = \mathrm{BN}(\mathcal{F}_{3\times3}(\mathrm{DOWN}(\boldsymbol{X}))) \tag{8}$$

$$\boldsymbol{X}_{\mathrm{GC}} = \sigma_{\mathrm{sig}}((\mathrm{UP}(\Delta\boldsymbol{X}) + \boldsymbol{X})) \tag{9}$$

where $\mathrm{DOWN}(\cdot)$ denotes the downsampling operation and $\mathrm{UP}(\cdot)$ represents the upsampling operation. Finally, $\boldsymbol{X}_{\mathrm{SR}}$ and $\boldsymbol{X}_{\mathrm{DC}}$ are concatenated, and the aggregated representation is dynamically modulated by $\boldsymbol{X}_{\mathrm{GC}}$. By incorporating the original input $\boldsymbol{X}$ to construct a residual connection, the structural integrity of key features in small objects is preserved. This approach effectively alleviates the feature degradation of dim and small objects within complex backgrounds in deep networks.

$$\boldsymbol{X}_{\mathrm{out}} = \boldsymbol{X}_{\mathrm{GC}} \odot \mathcal{F}_{1\times1}([\boldsymbol{X}_{\mathrm{SR}}, \boldsymbol{X}_{\mathrm{DC}}]) + \boldsymbol{X} \tag{10}$$

B. Cross-modal Gated Interaction Module

To address the semantic ambiguity that frequently occurs in traditional fusion methods under complex backgrounds, as well as the vulnerability of small objects to feature degradation caused by cross-modal mismatch, this paper proposes the CGIM, as illustrated in Fig. 3. To avoid the computational redundancy and feature dilution caused by conventional self-attention and Transformer-based fusion architectures, CGIM uses a linear gating unit for pixel-wise feature modulation. By utilizing a linear gating unit to induce post-interaction cross-modal saliency features as a guidance mechanism, the module executes pixel-wise dynamic gain modulation across the cross-modal fused feature streams.

In the nascent stage of fusion, the input visible features $\boldsymbol{F}_{\mathrm{vis}} \in \mathbb{R}^{C\times H\times W}$ and infrared features $\boldsymbol{F}_{\mathrm{ir}} \in \mathbb{R}^{C\times H\times W}$ are initially integrated via a concatenation operation to aggregate multimodal information. Subsequently, a 1×1 point-wise convolution is employed to project the feature space from $2C$ to $4C$. This dimensionality expansion is specifically engineered to induce nonlinear interactions between modalities and generate adaptive gating modulation map, thereby yielding an augmented feature $\boldsymbol{F}_{\mathrm{exp}} \in \mathbb{R}^{4C\times H\times W}$ enriched with cross-modal semantics:

$$\boldsymbol{F}_{\mathrm{exp}} = \mathcal{F}_{1\times1}^{\mathrm{PW}}([\boldsymbol{F}_{\mathrm{ir}}, \boldsymbol{F}_{\mathrm{vis}}]) \tag{11}$$

Considering the inevitable registration discrepancies between infrared and visible sensors, the multimodal features undergo local spatial interaction prior to deep fusion. The CGIM employs depth-wise convolution (DWConv) with a 3×3 kernel to aggregate local spatial context, where $\mathcal{F}_{3\times3}^{\mathrm{DW}}$ denotes the 3×3 depth-wise operator, thereby reducing the influence of minor spatial discrepancies in the feature domain. Following this local spatial interaction, the processed $4C$ features are decoupled along the channel dimension into two symmetric branches:

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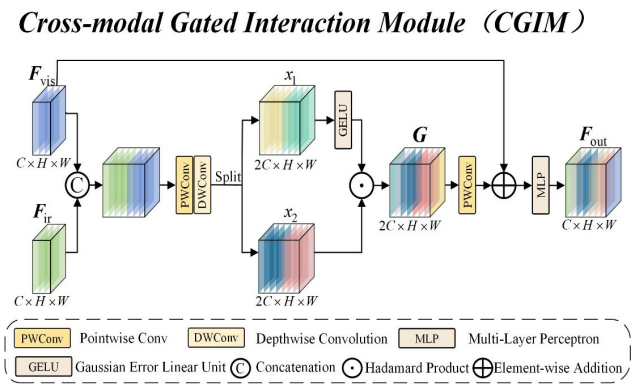


Fig. 3. Architecture of the CGIM.

$$(\boldsymbol{x}_1, \boldsymbol{x}_2) = \mathrm{Split}\left(\mathcal{F}_{3\times3}^{\mathrm{DW}}\left(\boldsymbol{F}_{\mathrm{exp}}\right)\right) \tag{12}$$

where $\boldsymbol{x}_1 \in \mathbb{R}^{2C\times H\times W}$ denotes the gating branch and $\boldsymbol{x}_2 \in \mathbb{R}^{2C\times H\times W}$ denotes the feature branch.

The theoretical core of the CGIM lies in the lightweight reconstruction of the dynamic weight allocation mechanism inherent in self-attention. Through the synergy of a dual-branch architecture and the GELU activation function, the CGIM emulates this process within a convolutional framework. Specifically, the gating branch $\boldsymbol{x}_1$, upon activation by GELU, generates a spatially adaptive gating modulation map $\boldsymbol{G}$ that executes a Hadamard product upon the feature branch $\boldsymbol{x}_2$.

$$\boldsymbol{G} = \mathrm{GELU}(\boldsymbol{x}_1) \odot \boldsymbol{x}_2 \tag{13}$$

This mechanism inherently functions as feature filtering guided by cross-modal semantic interaction. By learning the saliency distribution within the fused features, the model dynamically modulates the gain of the feature streams: high weights are assigned to target regions—such as those with prominent infrared thermal radiation or distinct visible textures—to enhance response intensity, whereas the gates tend to close in background interference zones to suppress the amplification of irrelevant features. To help retain low-level textures and edge information within the backbone, the CGIM incorporates residual reconstruction logic. The modulated features are reverted to dimension C via a 1×1 point-wise convolution and fused with the original visible features, effectively mitigating feature attenuation in deep networks and supporting semantic consistency of critical structural features. Finally, the fused features undergo deep refinement via a multilayer perceptron (MLP), which stabilizes the gradient distribution while bolstering the discriminative capability of the representations. This establishes a robust foundation for subsequent feature propagation across hierarchical stages.

$$\boldsymbol{F}_{\mathrm{out}} = \mathrm{MLP}\left(\mathcal{F}_{1\times1}^{\mathrm{PW}}\left(\boldsymbol{G}\right) + \boldsymbol{F}_{\mathrm{vis}}\right) \tag{14}$$

CGIM differs from the gated mechanisms used in GLU, GMU [64], and Gated Convolution [65]. GLU divides a transformed feature into a value branch and a sigmoid gate, while GMU independently transforms each modality and performs gated weighted fusion between modality-specific representations. Gated Convolution uses two parallel convolutions to generate a feature map and a spatial gate, generally followed by sigmoid modulation. In contrast, CGIM first concatenates multi-modal features, performs joint channel mixing and depth-wise spatial interaction, and then applies GELU-based multiplicative modulation with residual projection and MLP refinement.

C. Hierarchical Contextual Guidance Module

In cross-modal detection architectures, fused features often suffer from spatial detail dissipation during hierarchical propagation due to downsampling or repeated convolutions, which compromises the localization precision for tiny targets. To reactivate spatial contextual associations prior to the prediction stage, we propose the HCGM, as illustrated in Fig. 4. The central design of the HCGM lies in simulating a broad receptive field through hierarchical decoupling, thereby providing enhanced spatial modeling capabilities while mitigating the computational overhead typically associated with large-kernel convolutions.

Initially, the input feature $\boldsymbol{X} \in \mathbb{R}^{C\times H\times W}$ undergoes spatial integration via a spatial sampling operator $\mathcal{K}_{k\times k}(\cdot)$ to extract preliminary local structural features $\boldsymbol{X}_{\mathrm{local}}$. To capture long-range spatial dependencies, a sparse spatial sampling operator $\mathcal{K}_{k\times k}^{d}(\cdot)$ is employed; this further expands the receptive field through cross-pixel feature aggregation based on $\boldsymbol{X}_{\mathrm{local}}$, capturing global spatial features $\boldsymbol{X}_{\mathrm{global}}$:

$$\boldsymbol{X}_{\mathrm{local}} = \mathcal{K}_{k\times k}(\boldsymbol{X}) \tag{15}$$

$$\boldsymbol{X}_{\mathrm{global}} = \mathcal{K}_{k\times k}^{d}(\boldsymbol{X}_{\mathrm{local}}) \tag{16}$$

where $\mathcal{K}_{k\times k}(\cdot)$ and $\mathcal{K}_{k\times k}^{d}(\cdot)$ denote a local convolutional operator with a stride of 1 and a cross-pixel sampling operator with a sampling interval of d, respectively, whereas k denotes the size of the sampling kernel.

The HCGM deploys specialized sampling kernels tailored for features across different hierarchies. For shallow features, a large-scale sampling kernel $k = 23$ with a dilation rate $d=3$ is implemented to compensate for the insufficient receptive field, leveraging a broad sampling space to capture vital global context for diminutive objects. For intermediate features, a medium-scale configuration ($k=11$, $d=2$) ensures the fluid transmission of semantic information during downsampling. Regarding deep features, a small-scale kernel ($k = 7$, $d=2$) is adopted to refine boundary perception accuracy based on high-level semantic representation. The module aggregates the local detail-guided features $\boldsymbol{X}_{\mathrm{local}}$ and global context-enhanced features $\boldsymbol{X}_{\mathrm{global}}$ along the channel dimension to generate hybrid features. By preserving the original representations from different receptive field paths, a multi-scale representation space is constructed, effectively addressing the response intensity attenuation caused by feature sparsity during the deep propagation of small objects. These restructured features subsequently undergo cross-channel

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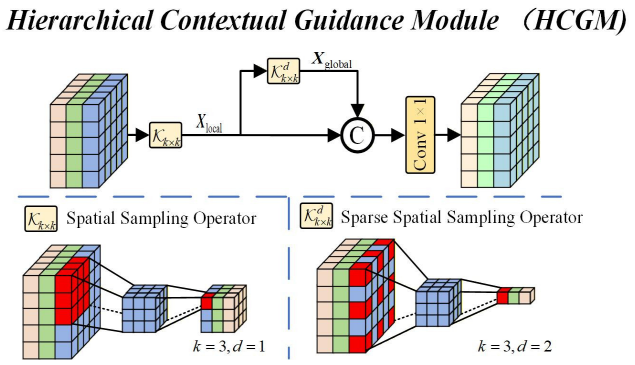


Fig. 4. Architecture of the HCGM.

fusion and dimensionality reduction via a 1×1 convolutional layer, yielding the final refined feature map $\boldsymbol{Y}$:

$$\boldsymbol{Y} = \mathcal{F}_{1\times1}([\boldsymbol{X}_{\text{local}}, \boldsymbol{X}_{\text{global}}]) \tag{17}$$

Through the spatial guidance of the HCGM, the resultant enhanced features provide the detection head with a highly discriminative spatial prior. This mechanism not only mitigates the information loss of small objects within deep feature hierarchies but also improves the representation consistency across multi-scale targets. Consequently, this design yields a measurable gain in detection performance amidst complex environmental backgrounds.

Three representative large-kernel mechanisms are considered: the 7×7 depth-wise convolution adopted in ConvNeXt [66], RepLKNet [67] large-kernel convolution with a small-kernel branch, and Large Kernel Attention [68] (LKA). ConvNeXt provides efficient feature extraction with low computational cost, RepLKNet captures long-range dependencies while preserving local details, and LKA adaptively emphasizes informative regions through an attention map. In contrast, HCGM combines local and dilated depth-wise convolution branches, concatenates their outputs, and fuses them with a 1×1 convolution, without relying on re-parameterization or an explicit attention map.

IV. Experiments

A. Datasets

To evaluate the performance of the proposed method, experiments were conducted on two publicly available and challenging aerial multi-scale small object detection datasets in the remote sensing field: DroneVehicle [39] and VEDAI [69]. For descriptive analysis in this study, we regard an object as small when both its normalized width and height are below 0.1 [70]. The majority of targets in the VEDAI dataset meet this definition, while only a small portion of targets in the DroneVehicle dataset satisfy this criterion, as shown in Fig. 5. To further validate the robust applicability across different scenarios and robustness of the model, this paper includes four additional multimodal datasets, LLVIP [71], M^{3}FD [72], MFAD [73], and KAIST [74], for supplementary experiments.

The DroneVehicle dataset is a large-scale vehicle detection

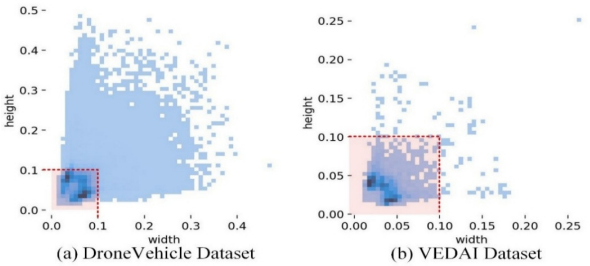


Fig. 5. Distribution of object sizes in the datasets.

benchmark captured from a UAV perspective. It contains 28,439 pairs of aligned visible and infrared images, covering five common vehicle categories: car, truck, bus, van, and freight car. The targets in these images are generally small and present challenges such as scale variation, low-light conditions, and dense occlusion. In this study, we strictly adhere to the data partitioning specified in the original literature, where the training, validation, and test sets contain 17,990, 1,469, and 8,980 image pairs, respectively.

The VEDAI dataset is extensively employed to evaluate detection algorithms for small and multi-category targets in aerial scenes. This dataset comprises 1,246 pairs of high-resolution visible and infrared images. Compared to DroneVehicle, VEDAI features a more diversified set of categories, encompassing 8 classes such as car, tractor, pickup, boat, and truck, which allows for a broader evaluation of the model's generalization ability in complex backgrounds. Regarding data partitioning, we strictly adhered to the official benchmarking protocol to ensure the fairness and comparability of the experimental results.

The LLVIP dataset is an aligned multispectral benchmark designed for pedestrian detection within traffic surveillance contexts. Comprising 15,488 synchronized visible-infrared image pairs at a resolution of 1280×1024, the dataset is partitioned into 12,025 training instances and 3,463 testing instances. While encompassing 24 nocturnal and 2 diurnal scenarios, the dataset ensures precise spatio-temporal registration across all modalities. Given that the majority of samples were captured under extreme low-light conditions and focus exclusively on the "person" category, LLVIP serves as a suitable benchmark standard for evaluating the robustness and adaptability of detection frameworks in degraded visual environments.

M^{3}FD is a multispectral object detection benchmark designed for complex environments. This dataset contains 4,200 strictly aligned visible and infrared image pairs with a resolution of 1024×768, covering various challenging all-weather scenes such as day, night, over-the-horizon, and smoke. A total of 33,603 object instances are annotated, encompassing six categories of traffic targets: pedestrian, car, bus, truck, lamp, and motorcycle. In the experiments, the training and testing sets consist of 3,360 and 840 image pairs, respectively, to evaluate the detection robustness of multimodal fusion models in complex scenarios.

The MFAD dataset is a visible-infrared benchmark for autonomous driving perception. It provides 12,370

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synchronized visible-infrared image pairs, including 9,897 pairs for training and 2,473 pairs for testing. The dataset contains six categories: car, bus, truck, pedestrian, e-bike rider, and cyclist. The images were collected in different driving environments, including roads, viaducts, tunnels, campuses, parks, residential areas, street blocks, and overpasses, under sunny, rainy, foggy, and hazy conditions. The variations in scene type, weather, and object appearance provide a comprehensive test of multimodal detection performance in practical driving environments.

The KAIST dataset is a public visible-infrared pedestrian detection benchmark collected in traffic scenes. It contains 11,215 paired visible and thermal images, including 8,963 pairs for training and 2,252 pairs for testing. The dataset covers pedestrians at near, medium, and far distances, with none, partial, and heavy occlusion conditions.

B. Experiment Settings

All experiments were conducted on the Ubuntu 20.04 operating system, utilizing a hardware environment equipped with a single NVIDIA GeForce RTX 3090 Ti GPU, and the proposed model was implemented using the PyTorch 2.5.1 and CUDA 12.1 deep learning frameworks. For all six datasets, all input images were uniformly resized to a resolution of 640×640 pixels before being fed into the network. The training process was set for 300 epochs with a batch size of 32. We employed stochastic gradient descent as the optimizer, featuring an initial learning rate of 0.01, a momentum factor of 0.937, and a weight decay coefficient of 0.0005. These hyperparameter configurations strictly adhere to the default training settings of YOLO11. To guarantee the consistency and fairness of the comparative experiments, other baseline methods were evaluated using their publicly available, reliable results. For open-source baselines, all training and testing were conducted under identical experimental parameters and hardware environments.

C. Evaluation Metrics

To comprehensively and intuitively evaluate the performance of the algorithm for small object detection in aerial scenarios, this study adopts two core categories of metrics: detection accuracy and computational efficiency. In terms of accuracy, precision (P), recall (R), and mean average precision (mAP) serve as the primary evaluative benchmarks. Precision and recall are quantified through the ratios of true positives (TP), false positives (FP), and false negatives (FN), while average precision is defined as the integral area under the precision-recall (P-R) curve. This analysis specifically focuses on mAP_{50} (the mean average precision across all categories at an intersection over union (IoU) threshold of 0.5) and $mAP_{50:95}$ (the average result calculated over an IoU range from 0.5 to 0.95 with an interval of 0.05), thereby providing a robust reflection of model performance under varying localization stringencies. Regarding efficiency, frames per second (FPS) is employed to assess real-time performance, encompassing the entire pipeline from image preprocessing and model inference to post-processing. A higher FPS value denotes superior detection efficiency and enhanced real-time responsiveness. Furthermore, to address the practical demands of embedded deployment, parameters (Params) and floating point operations (FLOPs) are introduced to quantify computational complexity. These metrics directly reflect the hardware resource consumption, serving as a critical reference for evaluating the feasibility of the algorithm on resource-constrained embedded devices.

D. Comparison With State-of-the-Art Methods

To validate the efficacy of the proposed MGIR-Net for aerial small object detection, rigorous comparative experiments were performed on the DroneVehicle, VEDAI, LLVIP, M^3FD, MFAD, and KAIST datasets. The suite of comparative algorithms encompasses dominant unimodal detection frameworks including RetinaNet, Faster R-CNN, RoITransformer, ReDet, FFCA-YOLO, and YOLO11-OBB. Additionally, advanced multimodal fusion architectures were evaluated, such as GLFDet, C^2Former, CFT, ICAFusion, MGFF, MSTD, DMM, MGMF, and CMFADet among others.

1) Comparisons on the DroneVehicle Dataset: Table I presents the performance of various object detection algorithms under unimodal and multimodal scenarios on the DroneVehicle dataset. Initial unimodal assessments reveal that the infrared modality consistently outperforms the visible modality. For instance, the performance of FFCA-YOLO improves from 74.5% with visible light input to 81.6% with infrared input, highlighting the value of thermal infrared information. By synergistically fusing visible and infrared data, the proposed MGIR-Net achieves a mAP_{50} of 85.5%. This result surpasses the best unimodal detection performance and outperforms all other multimodal fusion algorithms, including GLFDet (75.4%), C^2Former (74.3%), CFT (73.5%), ICAFusion (73.6%), MGFF (79.8%), MSTD (74.5%), DMM (79.4%), MGMF (80.1%), LCAFNet (76.2%), MCISFNet (82.9%) and CMFADet (83.4%). Specifically, MGIR-Net achieves a 2.1 % improvement over the previous state-of-the-art CMFADet (83.4%). Detailed category-specific analysis further confirms the effectiveness of MGIR-Net, featuring the highest results for trucks (84.8%), and vans (96.1%), alongside robust performance in challenging categories such as cars (97.2%) and freight cars (73.2%).

The visual comparative performance of various detection methods on the DroneVehicle dataset is illustrated in Fig. 6, where yellow and red circles denote missed detections and wrong classification, respectively. Competing algorithms are susceptible to classification misjudgments when processing semantic interference generated by vegetation and road textures. When visible light modalities fail or infrared thermal features are confounded with the background, particularly in deep shadow regions of architectural structures, the edge information of small objects becomes blended with the environment; consequently, traditional fusion architectures often suffer from feature dilution and semantic contamination, leading to missed detections of targets and the erroneous identification of similar background as targets. By leveraging feature reconstruction and dynamic cross-modal gain adjustment, MGIR-Net helps alleviate semantic

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TABLE I

COMPARISON OF DETECTION RESULTS ON THE DRONEVEHICLE DATASET

Method	Modality	Car	Freight	Truck	Bus	Van	mAP_{50}
RetinaNet[23]	Visible	67.5	13.7	28.2	62.1	19.2	38.1
FasterR-CNN[8]		67.9	26.3	38.6	66.9	23.2	44.6
RoITransformer[24]		68.2	29.0	44.1	70.5	27.7	47.9
ReDet[25]		69.5	31.5	47.8	77.4	29.1	51.1
YOLO11-OBB[11]		93.3	63.3	70.1	59.3	93.7	75.9
FFCA-YOLO[32]		92.1	56.5	69.6	93.7	60.8	74.5
RetinaNet[23]	Infrared	79.8	28.1	32.9	67.3	16.5	44.9
FasterR-CNN[8]		88.7	35.2	42.5	77.9	28.6	54.6
RoITransformer[24]		88.9	41.5	51.6	79.5	34.4	59.2
ReDet[25]		89.5	42.9	53.9	79.9	36.6	60.6
YOLO11-OBB[11]		98.3	66.9	77.7	95.2	61.2	79.9
FFCA-YOLO[32]		98.2	68.8	77.9	96.9	66.1	81.6
GLFDet[46]	Visible + Infrared	89.5	64.2	71.5	89.4	62.2	75.4
C^2Former[47]		90.2	64.4	68.3	89.9	58.5	74.3
CFT[48]		93.6	57.6	67.3	55.9	92.9	73.5
ICAFusion[49]		94.8	57	67.5	56.6	92	73.6
MGFF[50]		90.4	69.6	80.9	89.9	68.0	79.8
MSTD[51]		96.8	52.8	69.2	93.1	60.4	74.5
DMM[52]		90.4	68.2	79.8	89.9	68.6	79.4
MGMF[53]		91.4	**78.5**	70.1	91.1	69.4	80.1
CMFADet[54]		**98.6**	71.6	82.9	96.2	67.8	83.4
LCAFNet[55]		90.3	67.9	69.7	89.7	63.4	76.2
MCISFNet[56]		96.5	76.9	78.4	95.6	67.3	82.9
CCSFuse[57]		97.1	67.8	72.1	94.3	64.1	79.1
PAO-YOLO[58]		98.6	70.8	82.5	**97.1**	71.5	84.1
Baseline		96.7	71.2	79.9	72.4	95.4	83.1
MGIR-Net		97.2	73.2	**84.8**	76.1	**96.1**	**85.5**

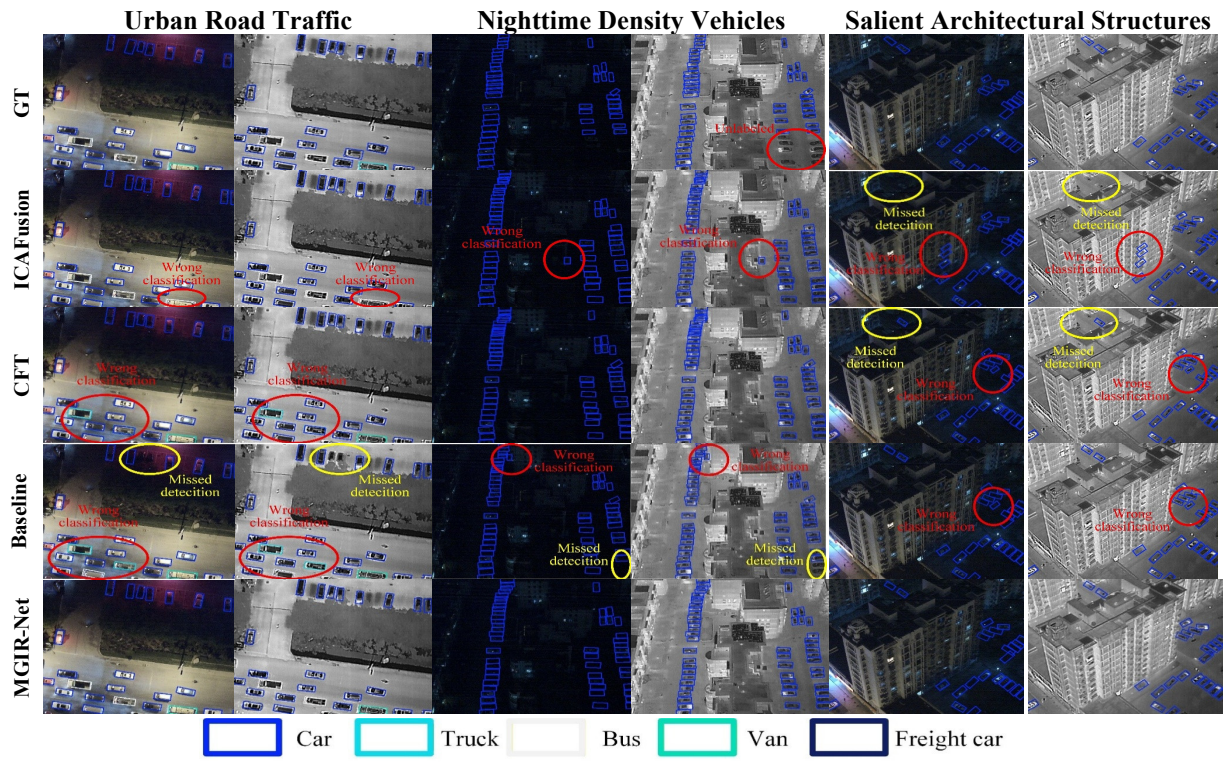


Fig. 6. Visual detection results on the DroneVehicle dataset.

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TABLE II
COMPARISON OF DETECTION RESULTS ON THE VEDAI DATASET

Method	Modality	Car	Pickup	Camping	Truck	Tractor	Boat	Van	Other	mAP_{50}
RetinaNet[23]	Visible	70.9	62.9	59.4	54.7	50.5	17.7	47.5	19.8	47.9
FasterRCNN[8]		83.2	80.3	70.2	63.2	66.2	60.1	82.9	39.8	68.2
RoITransformer[24]		86.3	80.6	70.3	56.6	71.8	69.4	71.2	40.1	68.3
ReDet[25]		77.3	71.5	73.6	56.1	64.7	56.7	60.2	42.8	62.9
YOLO11[11]		88.3	79.1	79.9	75.1	83.7	75.5	58.2	49.0	73.6
FFCA-YOLO[32]		89.3	86.7	76.9	74.7	78.5	49.7	74.6	51.0	72.7
RetinaNet[23]	Infrared	59.7	54.1	53.4	52.8	35.3	9.6	42.3	24.9	41.5
FasterRCNN[8]		82.7	78.6	70.7	60.1	51.6	62.9	76.7	43.1	65.8
RoITransformer[24]		84.7	74.4	70.2	57.7	57.2	60.8	75.9	25.6	63.3
ReDet[25]		76.1	70.7	71.2	51.7	51.9	46.9	64.3	28.3	57.6
YOLO11[11]		81.6	68.8	68.4	61.1	71.5	49.8	69.8	60.8	66.5
FFCA-YOLO[32]		85.5	86.3	**81.4**	74.8	68.1	59.9	86.2	29.8	71.5
GLFDet[46]	Visible + Infrared	87.6	84.6	74.9	67.8	75.5	62.1	74.3	50.8	72.2
C^2Former[47]		76.7	68.7	63.2	52.0	59.8	43.3	48.0	41.9	56.7
CFT[48]		81.8	79.3	79.8	77.5	**80.5**	48.6	60.0	71.4	72.4
ICAFusion[49]		84.4	82.5	73.4	76.2	59.9	53.0	81.9	69.5	72.6
DMM[52]		84.2	78.8	79.0	65.7	72.3	72.3	72.5	56.2	72.6
CMFADet[54]		88.2	78.2	77.6	**85.3**	79.6	62.8	77.1	61.4	76.3
MCISFNet[56]		89.4	**86.6**	78.6	68.0	77.6	66.7	72.4	51.6	73.9
Baseline		88.4	79.1	79.3	66.5	64.0	52.1	68.6	58.8	69.6
MGIR-Net		**90.1**	83.3	78.7	60.6	73.0	**77.5**	**86.6**	**78.9**	**78.6**

contamination and feature submergence of small objects in complex imaging scenes, thereby reducing the rates of missed detections and wrong classification.

2) Comparisons on the VEDAI Dataset: Table II presents the performance of various object detection algorithms on the VEDAI dataset under both unimodal and multimodal settings. Consistent with previous findings, most multimodal methods achieve better overall performance than unimodal detectors. The proposed MGIR-Net achieves an mAP_{50} of 78.6%, exceeding the best unimodal performance. It also outperforms established multimodal fusion algorithms, including GLFDet (72.2%), C^2Former (56.7%), CFT (72.4%), ICAFusion (72.6%), DMM (72.6%), MCISFNet (73.9%), and CMFADet (76.3%). Specifically, MGIR-Net improves the mAP_{50} by 9.0 % over the baseline (69.6%). Among the eight evaluated categories, MGIR-Net achieves the best performance in four categories: car (90.1%), boat (77.5%), van (86.6%), and other (78.9%). VEDAI also contains a broader range of categories and many extremely small objects.

The visual comparison of different detection methods on the VEDAI dataset is presented in Fig. 7, where the yellow and red circles denote missed detections and wrong classification, respectively. In agricultural scenes, background interference from bare ground and surface debris leads to missed detections and wrong classification. In urban scenes, architectural shadows and high-contrast road surfaces create substantial semantic ambiguity, making other algorithms less effective at extracting useful features from small objects and even resulting in the misidentification of rooftops as vehicles. In hybrid scenes characterized by dispersed targets and imbalanced modal information, extensive missed detections occur near building edges and bare-ground regions. VEDAI is therefore suitable for evaluating vehicle detection under small object and complex-background conditions. These results demonstrate that MGIR-Net reduces background interference and modal mismatch, leading to improved detection of diverse small objects.

3) Comparisons on the LLVIP Dataset: The experimental results presented in Table III indicate that multimodal fusion-based methods generally outperform single-modality approaches in detection accuracy, validating the value of bimodal information complementarity in complex scenarios. Compared to other state-of-the-art multimodal object detection methods, MGIR-Net achieves promising detection precision. Furthermore, MGIR-Net maintains stable detection performance when applied to the LLVIP non-aerial low-light dataset, achieving a mAP_{50} of 96.5%, which demonstrates the robust applicability across different scenarios and robustness of the proposed model.

As illustrated in Fig. 8, when subjected to challenges in complex nocturnal environments, including localized occlusions, thermal reflection artifacts, and vegetative obstruction, existing methods such as ICAFusion, CFT, and the baseline exhibit varying degrees of limitations, resulting in frequent missed detections or wrong classifications. In contrast, the detection results of the

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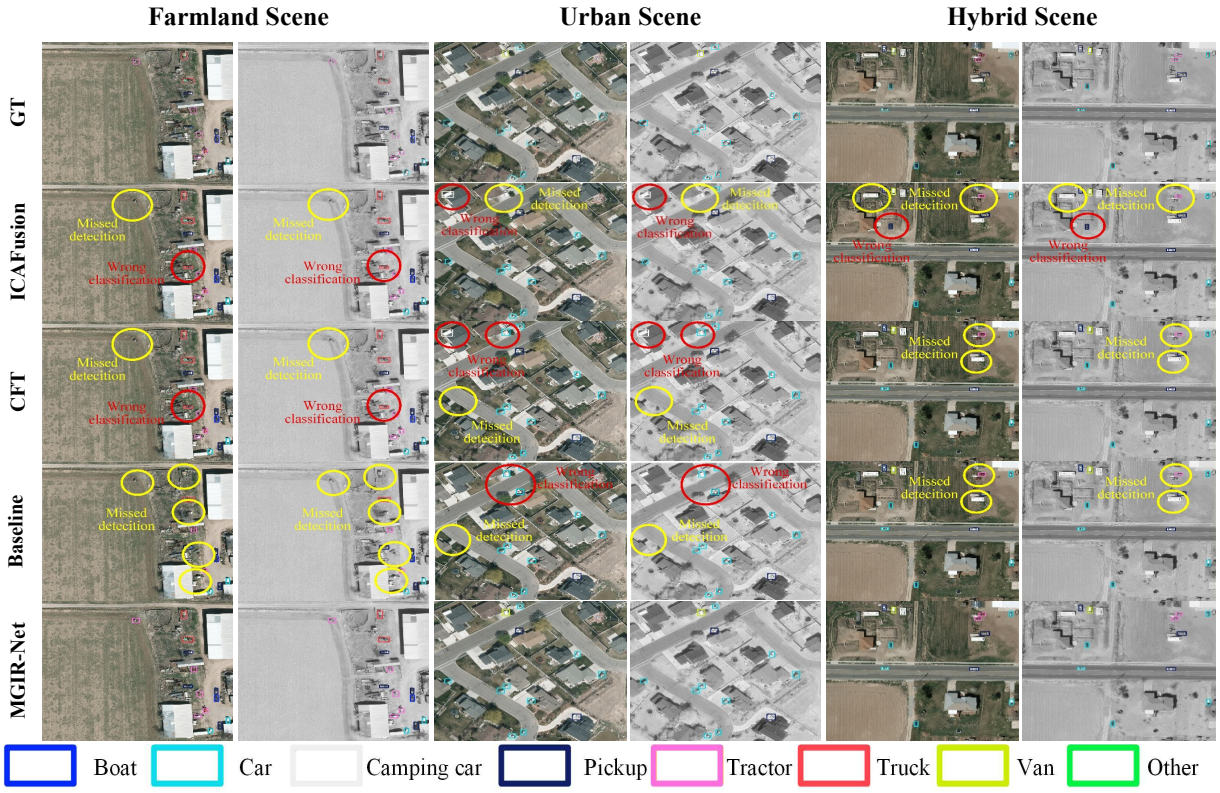


Fig. 7. Visual detection results on the VEDAI dataset.

TABLE III
COMPARISON OF DETECTION RESULTS ON THE LLVIP DATASET

Method	Modality	mAP_{50}
FasterR-CNN[8]	Visible	88.8
DETR[13]		89.5
Deformable DETR[15]		91.3
YOLO11[11]		89.0
FasterR-CNN[8]	Infrared	92.6
DETR[13]		93.0
Deformable DETR[15]		94.5
YOLO11[11]		94.7
Deformable DETR[15]	Visible + Infrared	95.2
GLFDet[46]		89.2
CFT[48]		95.5
ICAFusion[49]		95.4
Baseline		95.1
MGIR-Net		**96.5**

TABLE IV
COMPARISON OF DETECTION RESULTS ON THE M^3FD DATASET

Method	Modality	mAP_{50}
FasterR-CNN[8]	Visible	83.1
YOLO11[11]		74.0
FasterR-CNN[8]	Infrared	80.3
YOLO11[11]		72.3
CDDFuse[18]	Visible + Infrared	81.1
MMI-Det[45]		83.5
CFT[48]		85.1
ICAFusion[49]		85.0
Fusion-DETR[14]		86.9
Fusion-Mamba[60]		88.0
EI2Det[73]		86.2
MCINet[75]		89.3
Baseline		86.5
MGIR-Net		**90.5**

proposed MGIR-Net more closely match the ground truth, with fewer missed detections and wrong classification caused by spurious thermal responses. These results provide evidence of the performance of MGIR-Net in non-aerial conventional-scale object detection, while demonstrating the robust applicability across different scenarios and robustness of the model in handling other complex real-world scenarios.

4) Comparisons on the M^3FD Dataset: The experimental results in Table IV demonstrate that on the more generalized and complex M^3FD dataset, multimodal fusion methods generally outperform single-modality methods. MGIR- Net achieves a mAP_{50} of 90.5%, outperforming other mainstream multimodal methods in detection accuracy, thereby further validating the robust applicability across different scenarios and robustness of the proposed model.

5) Comparisons on the MFAD Dataset: The experimental results in Table V show that multimodal fusion-based methods generally outperform single-modality approaches on the diverse MFAD dataset. MGIR-Net achieves an mAP_{50} of 79.0%, providing evidence of its ability to exploit complementary visible-infrared information under varied

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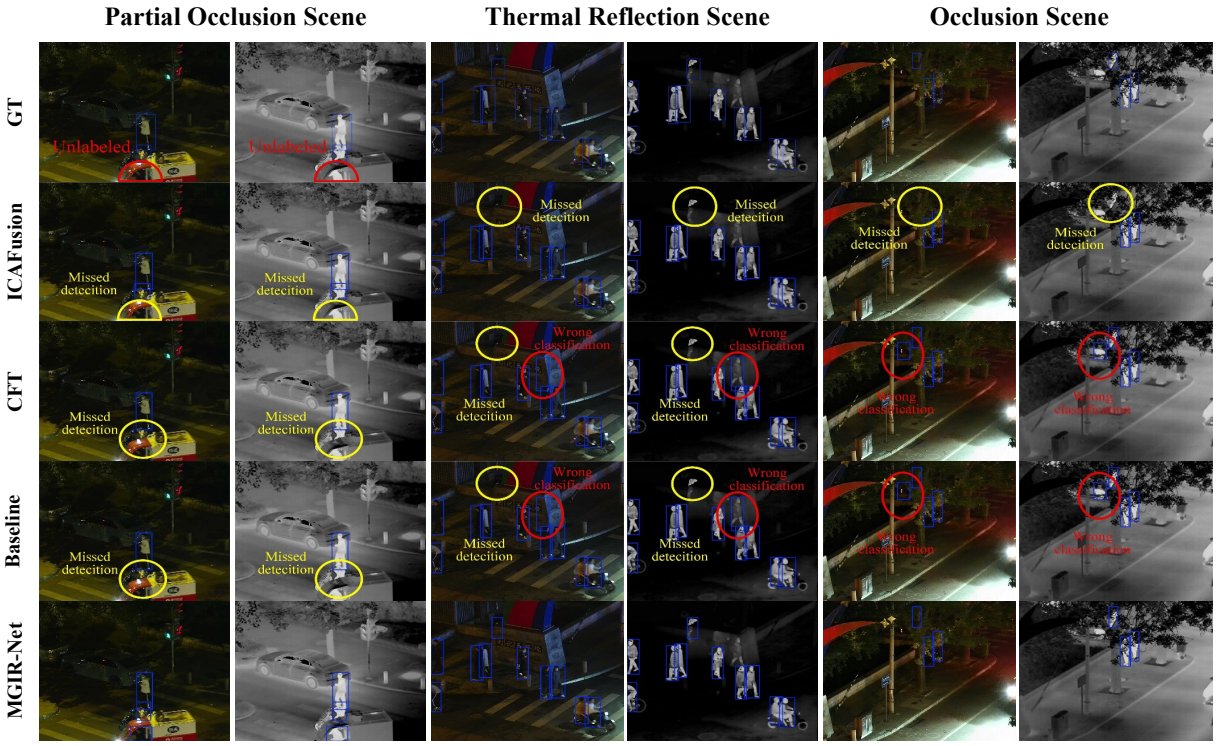


Fig. 8. Visual detection results on the LLVIP dataset.

TABLE V
COMPARISON OF DETECTION RESULTS ON THE MFAD DATASET

Method	Modality	mAP_{50}
YOLOv5[10]	Visible	74.9
YOLO11[11]		72.1
YOLOv5[10]	Infrared	70.0
YOLO11[11]		68.7
CFT[48]	Visible + Infrared	77.8
ICAFusion[49]		77.6
MMI-Det[45]		76.9
TarDAL[72]		69.8
TINet[76]		69.1
Baseline		76.0
MGIR-Net		**79.0**

TABLE VI
COMPARISON OF DETECTION RESULTS ON THE KAIST DATASET

Method	Modality	mAP_{50}
FasterR-CNN[8]	Visible	58.3
SSD[12]		48.2
RetinaNet[23]		57.7
YOLO11[11]		59.8
FasterR-CNN[8]	Infrared	68.6
SSD[12]		60.9
RetinaNet[23]		68.2
YOLO11[11]		70.3
MMTOD[77]	Visible + Infrared	70.7
CMDet[39]		68.4
GAFF[78]		67.1
CFT[48]		71.2
ICAFusion[49]		71.0
RISNet[79]		72.7
Baseline		71.4
MGIR-Net		**75.2**

driving scenes and weather conditions.

6) Comparisons on the KAIST Dataset: The experimental results in Table VI show that multimodal fusion-based methods generally outperform single-modality approaches on the KAIST pedestrian detection dataset. MGIR-Net achieves an mAP_{50} of 75.2%, indicating its generalization capability for pedestrian detection under different illumination, scale, and occlusion conditions.

E. Ablation Study

1) Quantitative Analysis of Functional Modules: As shown in Table VII, comprehensive ablation experiments are conducted on the DroneVehicle and VEDAI datasets using a dual-stream YOLO11 as the baseline. The baseline achieves mAP_{50} values of 83.1% and 69.6% on the two datasets, respectively. By reconstructing modality-specific and fused features, FRM preserves small object details and reduces interference from amplified background responses. It improves the mAP_{50} to 84.2% on DroneVehicle and 72.9% on VEDAI, corresponding to improvements of 1.1% and 3.3% over the baseline. CGIM performs cross-modal spatial interaction and adaptive gated modulation to regulate complementary VIS–IR information, achieving mAP_{50} values of 84.4% and 73.7%, with improvements of 1.3% and 4.1%, respectively. HCGM employs decoupled spatial sampling to

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TABLE VII
QUANTITATIVE ANALYSIS OF FUNCTIONAL MODULES

Method	FRM	CGIM	HCGM	GFLOPs	Params	DroneVehicle					VEDAI				
						P	R	mAP_{50}	$mAP_{50:95}$	FPS	P	R	mAP_{50}	$mAP_{50:95}$	FPS
Baseline	—	—	—	9.8	3.8 M	79.1	79.2	83.1	66.9	224.1	68.2	62.8	69.6	42.6	257.7
a	✓	—	—	15	5.7 M	78.2	81	84.2	67.6	70.1	73.9	66.7	72.9	49.1	72.6
b	—	✓	—	12.8	4.7 M	82.5	79.1	84.4	68.4	199.7	75.7	66.1	73.7	50.7	208.4
c	—	—	✓	9.7	3.7 M	79.4	78.3	83.5	67.5	229.1	65.9	72.4	72.1	47.7	199.8
d	✓	✓	—	18.6	6.8 M	81	80.3	85.1	69	68.1	72.5	66.5	74	49.6	70.4
e	✓	—	✓	15.6	5.8 M	80.8	79.6	84	67.9	68.3	68.2	66.8	73.5	50.5	70.7
f	—	✓	✓	13.3	4.8 M	82	78.7	84.4	68.2	191.2	68	70.3	74.2	49.3	180.2
g	✓	✓	✓	19.3	6.9 M	81.9	79.9	85.5	69.5	55.6	70.3	70.7	78.6	50.4	57.6

TABLE VII (CONTINUED)
QUANTITATIVE ANALYSIS OF FUNCTIONAL MODULES

Method	FRM	CGIM	HCGM	Inference Time	Peak GPU Memory
Baseline	—	—	—	3.79ms	114MB
a	✓	—	—	13.42ms	128MB
b	—	✓	—	4.78ms	138MB
c	—	—	✓	4.16ms	114MB
d	✓	✓	—	13.57ms	152MB
e	✓	—	✓	7.47ms	140MB
f	—	✓	✓	5.45ms	128MB
g	✓	✓	✓	17.01ms	152MB

capture broader contextual information, increasing the mAP_{50} to 83.5% and 72.1%, with gains of 0.4% and 2.5%.

Among the three modules, CGIM provides the largest standalone improvement on both datasets, indicating that gated cross-modal interaction is the primary source of the performance gain. FRM provides a comparable improvement through feature reconstruction, whereas HCGM offers an additional efficiency-oriented enhancement through contextual modeling. When all three modules are integrated, MGIR-Net achieves mAP_{50}/$mAP_{50:95}$ values of 85.5%/69.5% on DroneVehicle and 78.6%/50.4% on VEDAI. Compared with the baseline, these results represent improvements of 2.4%/2.6% and 9.0%/7.8%, respectively. The results show that the overall improvement arises mainly from gated cross-modal interaction and feature reconstruction, with HCGM providing complementary contextual enhancement.

The computational overhead of each module is also reported in Table VII. Compared with the baseline, FRM introduces 1.9M parameters and 5.2 GFLOPs, whereas CGIM adds 0.9M parameters and 3.0 GFLOPs. HCGM requires 3.7M parameters and 9.7 GFLOPs, slightly fewer than the baseline. The complete MGIR-Net contains 6.9M parameters and requires 19.3 GFLOPs. Inference time represents the average pure forward-processing latency for a single paired input with a batch size of 1. It is measured over 1000 runs after 200 warm-up iterations, excluding image preprocessing and post-processing, while peak GPU memory denotes the maximum GPU memory allocated during inference. Compared with the baseline, MGIR-Net increases the inference time from 3.79ms to 17.01ms and peak GPU memory from 114 MB to 152 MB. Nevertheless, it achieves 55.6 FPS on DroneVehicle and 57.6 FPS on VEDAI, remaining above the commonly used 30 FPS real-time threshold and demonstrating its applicability to real-time remote sensing detection.

TABLE VIII
ABLATION STUDY OF THE CGIM CORE COMPONENT

Method	Residual source	Interaction mode	mAP_{50}
Baseline	—	—	83.1
a	Infrared	Gated Interaction	83.0
b	Visible	Multiplicative Weighting	84.0
CGIM	Visible	Gated Interaction	**84.4**

2) Quantitative Analysis of the CGIM Core Component: The effectiveness of the core logical units within the CGIM was verified through quantitative indicators on the DroneVehicle dataset, focusing on the source of the cross-modal residual path and the interaction guidance mechanism for object detection performance, as shown in Table VIII. Specifically, "*Multiplicative Weighting*" represents a static fusion strategy that applies a direct element-wise multiplication between cross-modal features or weights, which lacks non-linear selection. In contrast, "*Gated Interaction*" introduces a non-linear modulation mechanism that utilizes a dual-branch structure paired with a GELU activation function to generate a spatially adaptive gating modulation map, performing a Hadamard product with the feature branch for pixel-level dynamic gain control. Experiments demonstrate that using the saliency features after bimodal fusion as a gate guidance mechanism outperforms the conventional multiplication weighting. Furthermore, introducing visible light features as the residual path achieves the best performance among the tested variants, as it helps preserve low-level texture and edge information, thereby effectively mitigating the degradation of detailed features in deep networks.

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TABLE IX
PERFORMANCE COMPARISON BETWEEN CGIM AND TRANSFORMER-BASED FUSION MODULES

Method	GFLOPs	Params	mAP_{50}	FPS
Baseline	9.8	3.8 M	83.1	224.1
TransformerFusionBlock[49]	14.7	7.1 M	83.7	162.6
CrossTransformerFusion[48]	15.5	6.2 M	83.5	140.4
Our CGIM	12.8	4.7 M	**84.4**	199.7

TABLE X
ROBUSTNESS TO CROSS-MODAL MISALIGNMENT ON THE DRONEVEHICLE DATASET (mAP_{50} / $mAP_{50:95}$)

Method	0 px	1 px	3 px	5 px	7 px
Baseline	83.1/66.9	81.8/63.4	81.6/65.2	81.2/62.3	79.5/61.2
ICAFusion	73.6/50.3	72.3/49.1	71.2/49.5	71.0/47.8	70.9/44.1
CFT	73.5/51.5	72.4/48.6	71.1/47.3	71.1/48.2	70.5/47.3
CGIM	84.4/68.4	83.7/66.8	83.3/65.9	83.1/64.4	82.5/62.2
MGIR-Net	85.5/69.5	84.4/67.9	84.1/65.4	84.1/63.5	83.6/63.2

3) Comparison of CGIM and Transformer-based Fusion Modules: According to the comparative results on the DroneVehicle dataset in Table IX, the CGIM demonstrates superior performance in cross-modal feature fusion. In contrast to computationally intensive Transformer-based fusion modules, the CGIM achieves comparable or even superior precision gains (mAP_{50} increased by 1.3%) while reducing computational overhead, adding a mere 3 GFLOPs to the baseline framework. The experimental data demonstrate that the CGIM effectively captures critical cross-modal interactions.

4) Robustness to Cross-Modal Misalignment: To investigate the influence of cross-modal misalignment, we construct different perturbation settings by horizontally shifting the visible images to the left by 1, 3, 5, and 7 pixels relative to the infrared images. The 0-pixel setting represents the original aligned image pairs. For a fair comparison, all methods are retrained and evaluated under each corresponding shift setting using the same training configuration. As shown in Table X, the performance of all methods generally decreases as the translation magnitude increases, indicating that cross-modal misalignment adversely affects detection performance. Nevertheless, CGIM and MGIR-Net consistently maintain higher absolute performance than the baseline, ICAFusion, and CFT. Under the 7-pixel shift, MGIR-Net achieves 83.6% mAP_{50} and 63.2% $mAP_{50:95}$, exceeding the baseline by 4.1% and 2.0%, respectively. CGIM also maintains competitive performance, reaching 82.5% mAP_{50} and 62.2% $mAP_{50:95}$. From the aligned setting to the 7-pixel shift, the mAP_{50} values of CGIM and MGIR-Net decrease by only 1.9%, whereas the baseline, ICAFusion, and CFT decrease by 3.6%, 2.7%, and 3.0% respectively. These results suggest that the gated interaction mechanism improves adaptation to moderate cross-modal misalignment, particularly in terms of mAP50.

TABLE XI
COMPARISON OF CGIM, GLU, GMU, AND GATED CONVOLUTION ON THE DRONEVEHICLE DATASET

Method	GFLOPs	Params	mAP_{50}	FPS
Baseline	9.8	3.8 M	83.1	224.1
GLU[62]	13.02	4.7 M	83.7	209.3
GMU[64]	16.40	5.8 M	83.1	208.4
Gated Convolution[65]	21.43	7.4 M	84.3	208.4
CGIM	12.8	4.7 M	**84.4**	199.7

TABLE XII
ABLATION STUDY OF THE FRM CORE COMPONENT

Method	LSR	DSC	GBC	GFLOPs	Params	mAP_{50}	FPS
Baseline	—	—	—	9.8	3.8 M	83.1	224.1
a	✓	—	—	15.05	5.3 M	83.7	90.2
b	—	✓	—	15.04	5.2 M	83.3	78.6
c	—	—	✓	14	5.1 M	83.7	150.3
d	✓	✓	—	15.4	5.4 M	83.8	94.7
e	✓	—	✓	14.92	5.5 M	84.0	121.9
f	—	✓	✓	14.92	5.5 M	83.9	125.9
g	✓	✓	✓	15	5.7 M	**84.2**	70.1

5) Comparison of CGIM With Existing Gated Fusion Modules: Table XI compares CGIM with GLU, GMU, and Gated Convolution on the DroneVehicle dataset. The proposed CGIM achieves an mAP_{50} of 84.4%, which is 0.7%, 1.3%, and 0.1% higher than GLU, GMU, and Gated Convolution, respectively. GLU mainly performs value-gate modulation, GMU focuses on modality-wise weighted fusion, and Gated Convolution generates a feature map and a spatial gate, but these modules provide limited joint modeling of cross-modal spatial complementarity. In contrast, CGIM combines depth-wise spatial interaction, GELU-based adaptive gating, and residual refinement to dynamically regulate visible-infrared features. This design alleviates modal discrepancy, background interference, and small object feature attenuation, resulting in more effective multimodal fusion and higher detection accuracy.

6) Quantitative Analysis of the FRM Core Component: To verify the effectiveness of each core component within the proposed FRM, we conduct a comprehensive ablation study on the DroneVehicle dataset, as summarized in Table XII. Compared to the baseline model (83.1% mAP_{50}), introducing individual components yields distinct performance improvements. Specifically, incorporating the LSR module boosts the mAP_{50} to 83.7%, validating that reinforcing local visual saliency can improve weak-target responses in complex background clutter. Meanwhile, applying the DSC and GBC modules increases the detection accuracy to 83.3% and 83.7%, respectively, which demonstrates the efficacy of stripping incoherent semantic interference and correcting global feature bias. Importantly, when all three sub-modules are integrated,

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TABLE XIII
COMPARISON OF FRM, CBAM, CA, AND GLU ON THE DRONEVEHICLE DATASET

Method	GFLOPs	Params	mAP_{50}	FPS
Baseline	9.8	3.8 M	83.1	224.1
CBAM[63]	11.93	4.3 M	83.7	109.1
CA[61]	11.75	4.3 M	82.6	81.0
GLU[62]	11.73	4.3 M	83.3	126.7
FRM	15	5.7 M	**84.2**	70.1

TABLE XIV
PERFORMANCE COMPARISON OF DIFFERENT SAMPLING KERNEL SIZES IN HCGM

Method	GFLOPs	Params	mAP_{50}	FPS
Baseline	9.8	3.8 M	83.1	224.1
$\{7,11,23\}$	9.71	3.75 M	82.6	235.6
$\{35,23,11\}$	10.03	3.76 M	83.0	229.9
$\{41,35,23\}$	10.10	3.78 M	83.3	237.1
$\{23,11,7\}$	9.75	3.73 M	**83.5**	229.1

our full FRM architecture achieves the highest performance of 84.2% mAP_{50}. This clear cumulative gain demonstrates the complementary nature of the components inside FRM.

7) Comparison of FRM With Other Modules: Table XIII compares FRM with CBAM, CA, and GLU under the same DroneVehicle detection architecture. FRM achieves the highest mAP_{50} of 84.2%, outperforming CBAM, CA, and GLU by 0.5%, 1.6%, and 0.9%, respectively. CBAM, CA, and GLU mainly perform feature reweighting or simple gated modulation, without explicitly reconstructing degraded features or suppressing directional and global background interference. In contrast, FRM combines LSR, DSC, and GBC to enhance local target saliency, calibrate directional semantic responses, and correct global feature bias. These complementary operations effectively address small object feature degradation and background interference, leading to better detection performance.

8) Quantitative Analysis of HCGM Sampling Kernel Size: To verify the impact of the hierarchical sampling kernel sizes configured for different feature levels in the proposed HCGM, quantitative comparison experiments were conducted on the DroneVehicle dataset, as shown in Table XIV. The experimental results indicate that the HCGM module achieves the best performance among the evaluated configurations with the sampling kernel combination configured as $\{23,11,7\}$.

9) Comparison of HCGM With Existing Large-Kernel Modules: Table XV compares HCGM with ConvNeXt, RepLKNet, and LKA on the DroneVehicle dataset. HCGM achieves an mAP_{50} of 83.5%, outperforming ConvNeXt, RepLKNet, and LKA by 0.5%, 0.4%, and 0.4%, respectively. ConvNeXt, RepLKNet, and LKA are designed to enlarge the receptive field and model broader spatial dependencies, but their general-purpose structures do not explicitly adapt the receptive field to the hierarchical spatial characteristics of small object. In contrast, HCGM combines local and dilated spatial sampling branches with hierarchical kernel configurations to preserve fine-grained target details and capture useful contextual relationships. This design better addresses small object feature loss and complex-background interference, resulting in improved detection performance with lower computational cost.

TABLE XV
COMPARISON OF HCGM WITH CONVNEXT, REPLKNET, AND LKA ON THE DRONEVEHICLE DATASET

Method	GFLOPs	Params	mAP_{50}	FPS
Baseline	9.8	3.8 M	83.1	224.1
ConvNeXt[66]	11.23	4.5 M	83.0	204.9
RepLKNet[67]	11.74	4.6 M	83.1	216.4
LKA[68]	10.28	4.0 M	83.1	209.8
HCGM	9.7	3.7 M	**83.5**	229.1

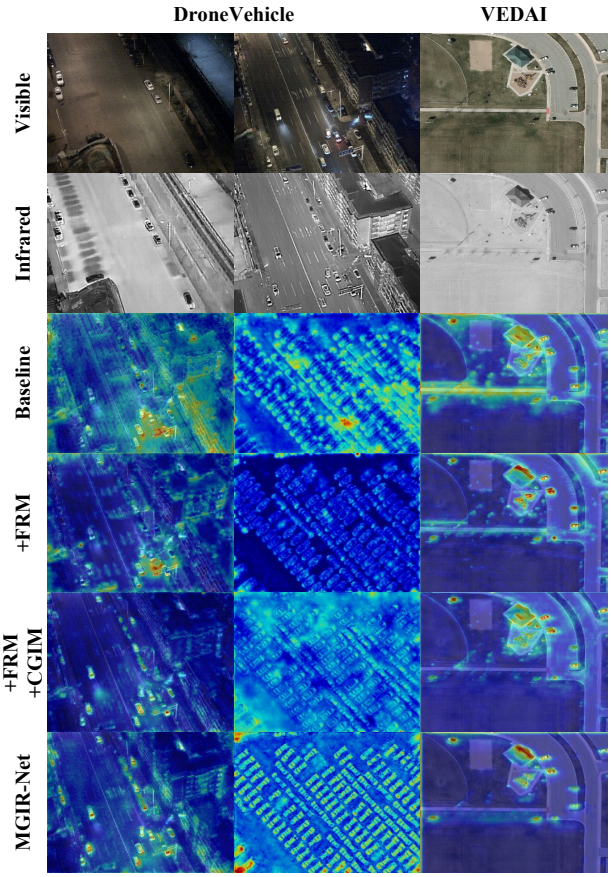


Fig. 9. Feature heatmap visualization comparison.

10) Comparison of Feature Map Visualizations: To intuitively verify the performance of the proposed algorithm in terms of feature representation, this section presents a visual comparison of the feature heatmaps generated at different

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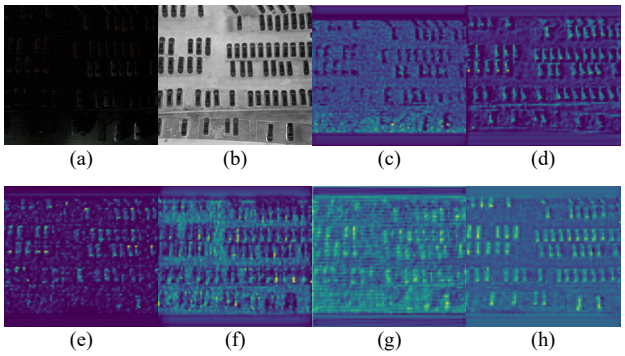


Fig. 10. Visual comparison of feature maps for cross-modal fusion modules. (a) Visible image. (b) Infrared image. (c) Visible feature map. (d) Infrared feature map. (e) Baseline. (f) ICAFusion's fusion module: TransformerFusionBlock. (g) CFT's fusion module: CrossTransformerFusion. (h) Our MGIR-Net's fusion module: CGIM.

stages of the model, as depicted in Fig. 9. In complex urban scenes or low-light environments, the feature response distribution of the baseline model is uneven. Strong activation responses are generated at road edges, architectural shadows, and extensive non-target areas, while the target contours remain blurred. Following FRM, large-scale background responses are suppressed, and small-target responses become clearer and more concentrated on vehicle regions. After CGIM is introduced, background clutter is further reduced and target boundaries become sharper. Within the complete MGIR-Net, the highlighted regions align more closely with the target contours.

To further verify the efficacy of the CGIM in cross-modal feature fusion, a visual comparison of feature maps under different fusion strategies is presented in Fig. 10. Although conventional Transformer-based fusion mechanisms, such as ICAFusion and CFT, capture some targets, modal misalignment leads to spatial positional deviations; consequently, the feature response distribution is scattered and the edge contours are indistinct. In contrast, the CGIM not only reduces redundant background responses during modal fusion but also enhances target responses through inter-modal interactive fusion, resulting in sharp target contours and a cleaner background region.

V. Conclusion

In this article, we propose MGIR-Net, a lightweight visible–infrared object detection framework tailored for UAV remote sensing imagery under complex backgrounds and challenging imaging conditions. To address small object feature degradation, cross-modal semantic interference, and spatial information loss during deep feature propagation, MGIR-Net introduces an FRM for modality-specific and post-fusion feature reconstruction and background interference suppression, a CGIM for adaptive gated visible–infrared interaction, and an HCGM for hierarchical contextual guidance before prediction. Extensive experiments on DroneVehicle, VEDAI, LLVIP, M^3FD, MFAD, and KAIST demonstrate that MGIR-Net achieves competitive detection performance across both aerial remote sensing and general visible–infrared benchmarks. Overall, this work highlights the importance of preserving discriminative small object features, suppressing unreliable modal responses, and exploiting lightweight contextual guidance in multimodal object detection.

Although MGIR-Net achieves improved performance across multiple datasets, several limitations remain. First, CGIM performs gated interaction based on information jointly extracted from visible and infrared features. When one modality is severely degraded or unavailable, unreliable responses may still affect the learned gating weights and weaken complementary fusion. Future work will introduce modality-quality estimation and degradation-aware gating to dynamically adjust the contribution of each modality under adverse illumination, sensor noise, missing-modality, and spatial misalignment conditions. Second, although the experiments cover several visible–infrared datasets, the model is still trained and evaluated within individual datasets, and its transferability across different sensors, geographic regions, weather conditions, and target distributions has not been fully examined. Large-scale remote sensing foundation models may also be incorporated to improve transferable multimodal representations and reduce dependence on dataset-specific training. Finally, the dual-stream backbone and multi-stage feature interaction increase inference time and memory consumption despite the lightweight design of HCGM. Future work will explore selective cross-modal interaction, shared feature extraction, model pruning, and knowledge distillation to further improve deployment efficiency on resource-constrained UAV platforms.

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